

Emirhan Yıldırım

Founder & Research Architect, Invariant Computing Program

Founder & Project Director, RARI • Founder / Independent Research Lead, Humanization Tech • Systems Architect, Invariant OS (in formation) • Founder / Technical Architect, Geometrodynamics Computing (in formation)

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PROFILE

Independent researcher, founder, and systems designer building a long-horizon programme around observer-aware computation: how information should represent its observer, transformation/filter, uncertainty, provenance, recoverability, equivalence classes, and structures that remain invariant across observers or transformations.

The current architecture places the Invariant Computing Program above the neuroscience-first Reality & Agency Research Initiative (RARI). RARI serves as the foundational scientific engine; Invariant OS is the design-stage computing-systems implementation path; Geometrodynamics Computing is the design-stage spatial/world-representation branch; Humanization Tech remains the separate biomedical/AI commercialization vehicle; and Elaveteos Studio remains an independent simulation/game-development brand.

CURRENT RESEARCH POSITION

- Current 15-day RARI falsification programme is in progress. Day-1 and Day-2 authoritative synthetic runs generated data trajectories; P1-P5 scores were derived afterward through frozen post-run analysis rather than emitted directly by the runs.
- Repeated cross-day signal: controlled perturbation expanded local identifiable rank from 8 to 17 in both completed synthetic phases; effect magnitude on predictive and uncertainty metrics was protocol-dependent.
- Current reconstruction bridge remains unresolved: the present observer-specific inverse/recovery procedure failed to reduce cross-observer dispersion in both synthetic phases. This is treated as a localized falsification target, not as evidence that observer-independent structure is impossible.
- Next phase: prospective real-biology, cross-species and cross-modality validation intended to test which mechanisms generalize beyond the synthetic system.

Status discipline: Design-stage programmes, planned ventures, preprints, technical disclosures, and in-progress experiments are labeled by status and are not presented as completed companies, granted patents, or validated scientific laws.

ROLE ARCHITECTURE

Default: Founder & lead architect across the Invariant Computing research and venture ecosystem.
SACRA: Assistant Research Lead (Program Lead: Ayşe Gülnaz).
Hoshi no Umi Studio / Purpura Yakamoz: CTO (Studio Head: Ayşe Gülnaz).
Ayşe Gülnaz: core co-builder across the wider ecosystem; bioengineering & human-systems remit.

WEB PRESENCE

Personal portfolio	emirhanyildirim-cloud.github.io
GitHub	github.com/emirhanyildirim-cloud
Ecosystem	invariantcomputinglab.org



PORTFOLIO



ECOSYSTEM

RESEARCH PROGRAM ARCHITECTURE

Foundational research only. Venture and implementation programmes are separated onto the follo

Research hub: research.invariantcomputinglab.org



Invariant Computing Program

Umbrella intellectual, scientific, and engineering programme

Long-horizon objective: develop a computing foundation in which information is explicitly conditioned on observer, transformation, uncertainty, provenance, recoverability and cross-observer invariants. The four foundational research programmes below remain conceptually distinct from downstream ventures and product systems.



RARI

Reality & Agency Research Initiative

Foundational research on observer/filter transformations, perturbation, invariants, agency, identity, identifiability and recoverability. Current work is organized around falsifiable perturbation experiments and observer-specific structure.

LEADERSHIP

Emirhan Yıldırım - Founder & Research Lead
Ayşe Günnaz - Co-builder, Bioengineering & Human Systems

rari.invariantcomputinglab.org



Live programme links: click a domain or scan its QR.

Research Programme I



OCIL / OCIR

Observer-Conditioned Language / Representation

Programme-level language, representation and reference-semantics research for observer-conditioned information. It defines scientific constraints and formal interfaces that downstream systems may implement without absorbing the research programme itself.

LEADERSHIP

Emirhan Yıldırım - Founder & Research Architect
Ayşe Günnaz - Co-builder, Biological Semantics

ocil-ocir.invariantcomputinglab.org



Research Programme II



AFC

Annihilation-Free Computing

New foundational research programme testing whether computation can be constructed from relational distinctions, transformations and preserved invariants without treating zero as a primitive computational state.

LEADERSHIP

Emirhan Yıldırım - Founder & Research Lead
Ayşe Günnaz - Co-builder, Cross-disciplinary Research

annihilation-free-computing.invariantcomputinglab.org



Research Programme III - NEW



SACRA

Symbolic Authority, Conversion & Ritual Agency

Foundational research programme examining how symbolic authority, ritual practice, repentance or conversion, tradition and perceived external agency reorganize identity and state transitions. Emphasis is placed on hysteresis, lock-in and effectively irreversible change.

LEADERSHIP

Ayşe Günnaz - Program Lead
Emirhan Yıldırım - Assistant Research Lead

sacra.invariantcomputinglab.org



Research Programme IV - NEW

PROGRAMME BOUNDARY

- Research side: RARI, OCIL/OCIR, AFC and SACRA.
- Venture / implementation side: Invariant OS, Geometrodynamic Computing, Humanization Tech, ElevateOS Studio, ATMER, Aurelis Space, Lutetia Retreats, Vectaris Mobility, Hoshi no Umi Studio and Purpura Yakamoz.
- Geometrodynamic Computing is explicitly treated as a venture / implementation programme here, not as a foundational research programme.

VENTURE & IMPLEMENTATION PROGRAMMES

Commercialization, systems, infrastructure and creative technology programmes kept separate from

Venture hub: ventures.invariantcomputinglab.org

Architecture rule

These programmes may implement, commercialize or experimentally apply outputs of the research layer. They are not counted as foundational research programmes.



Invariant OS

Computing systems branch

Design-stage implementation layer for language tooling, compiler/IR, runtime, GNU/Linux-based research OS components, storage/networking and later architecture/ISA work where technically justified.

invariant-os.invariantcomputinglab.org



ROLE

Emirhan - Systems Architect & Venture Lead
Ayşe - Co-builder, Human/Biological Interface

Systems implementation venture



Geometrodynamic Computing

Spatial / world-representation venture programme

Venture and technical programme for 3D/4D world representation, manifold/state-space systems, coordinate transforms, sensor/world fusion, mixed reality and persistent-world systems.

geometrodynamic-computing.invariantcomputinglab.org



ROLE

Emirhan - Founder & Technical Architect
Ayşe - Co-builder, Human/Embodied Systems

Venture programme



Humanization Tech

Applied biomedical / AI commercialization

Execution and commercialization vehicle for biomedical representation, bounded AI, trustworthy data infrastructure, rehabilitation technology and deep-tech prototyping.

humanization-tech.invariantcomputinglab.org



ROLE

Emirhan - Founder & Independent Research Lead
Ayşe - Co-builder, Biological Semantics & Documentation

Active venture



ElevateOS Studio

Games / simulation / creative technology

Independent game-development and simulation brand for Prometheus One World / GameKey, persistent-world architecture, Unreal prototyping and creator-led interactive systems.

elevateos-studio.invariantcomputinglab.org



ROLE

Emirhan - Founder & Technical/Creative Systems Lead
Ayşe - Co-builder, Human-Centered Experience

Active brand



ATMER

Adaptive AI Execution Infrastructure

Vendor-neutral routing and control layer for model selection, reasoning budgets, agent/workflow execution, compute placement and verification.

atmer.invariantcomputinglab.org



ROLE

Emirhan - Founder & AI Systems Lead
Ayşe - Co-builder, Human/Biological Evaluation

AI infrastructure venture



Aurelis Space

Inner Solar System Technologies

Space-technology application venture for autonomous onboard computing biosystems, Venus systems and progressively lower-perihelion solar-proximity

aurelis-space.invariantcomputinglab.org



ROLE

Emirhan - Founder & Technical Architect
Ayşe - Co-builder, Bioengineering & Life Systems

Application venture - in formation

VENTURE & IMPLEMENTATION PROGRAMMES - CONTINUED

Additional independent cross-ecosystem ventures under the Invariant Computing Program.

Sibling venture rule

Lutetia Retreats, Vectaris Mobility, Hoshi no Umi Studio and Purpura Yakamoz are independent venture structures within the implementation layer. They may integrate technologies and research outputs from sibling programmes while retaining distinct brands, products, operations and revenue models.



Lutetia Retreats

Experiential Hospitality / Modular Architecture

Independent hospitality and spatial-experience venture under the Invariant Computing Program. Develops modular, design-led retreat environments and serves as a cross-ecosystem physical deployment layer for selected technologies, interactive systems and research outputs produced across the wider Invariant Computing venture network.

LEADERSHIP

Emirhan - Founder & Technology Architect
Ayşe - Co-builder, Human-Centered Systems

CORE SCOPE

Hospitality / modular architecture / experiential tourism / nature-integrated accommodation

Independent hospitality & spatial-experience venture



Vectaris Mobility

Transportation & Mobility Systems Venture

Independent transportation venture under the Invariant Computing Program, developing vehicles, autonomous mobility systems, intelligent transportation infrastructure and multimodal transport technologies. Designed as a cross-ecosystem implementation platform spanning automotive, public transport, logistics, rail, marine and emerging mobility systems.

LEADERSHIP

Emirhan - Founder & Mobility Systems Architect
Ayşe - Co-builder, Human Factors & Bioengineering

CORE SCOPE

Vehicles / autonomous mobility / public & shared transport / logistics / rail / marine / mobility infrastructure

Independent transportation / mobility venture



Hoshi no Umi Studio

Anime / Webtoon / Narrative Universe Studio

Independent narrative and visual-IP venture under the Invariant Computing Program. Studio Head: Ayşe Günnaz. CTO: Emirhan Yıldırım. Develops anime-inspired story worlds, serialized webtoon properties and character-driven transmedia universes with strong visual identity, immersive worldbuilding and emotionally resonant storytelling.

LEADERSHIP

Ayşe Günnaz - Studio Head
Emirhan Yıldırım - CTO

CORE SCOPE

Webtoon / anime development / narrative IP / character worlds / visual storytelling

Independent anime / webtoon studio venture



Purpura Yakamoz

YouTube / Creator Media / Philosophical Storytelling

Independent creator-media venture under the Invariant Computing Program. Studio Head: Ayşe Günnaz. CTO: Emirhan Yıldırım. Develops YouTube content, reflective storytelling, media essays and emotionally resonant digital narratives oriented toward a brighter future.

LEADERSHIP

Ayşe Günnaz - Studio Head
Emirhan Yıldırım - CTO

CORE SCOPE

YouTube channel / media essays / storytelling / creator media / digital content

Independent creator-media / content venture

Cross-ecosystem integration

All four ventures may selectively draw on Invariant OS, Geometrodynamics Computing, Humanization Tech, ElevateOS Studio, ATMER and Aurelis Space, together with applicable outputs from RARI, OCIL / OCIR, AFC and SACRA. This relationship is collaborative rather than hierarchical: neither venture is a subsidiary of another ecosystem venture.

CURRENT FOUNDATIONAL RESEARCH PROGRAMMES

Research programmes only; venture and implementation programmes are separated on the preceding architecture page.

Invariant Computing Program - umbrella architecture

The umbrella programme defines the observer-conditioned information model, invariance criteria, provenance, recoverability, uncertainty and cross-observer comparison framework. The four research programmes below develop distinct scientific layers inside that architecture; downstream systems and ventures are treated separately.

1. RARI - REALITY & AGENCY RESEARCH INITIATIVE

Research Programme I - observer, agency, identity and falsification

RARI is the foundational science programme for the observer/filter side of Invariant Computing. It studies how source or world states are transformed by observer-specific filters, which structures remain identifiable under perturbation, and which claims about identity or agency survive falsification.

Current work uses controlled perturbation and synthetic-to-biological validation paths rather than treating reconstruction success as assumed. The central question is which observer-conditioned structures recur across perturbation classes, species and modalities.

- Core themes: observer/filter transformations, invariants, perturbation, identifiability, recoverability, agency and preserved identity.
- Current scientific boundary: observer-independent reconstruction is not treated as validated merely because local identifiability improves.

2. OCIL / OCIR - OBSERVER-CONDITIONED LANGUAGE / REPRESENTATION

Research Programme II - formal language, representation and reference semantics

OCIL/OCIR develops the programme-level language and intermediate-representation layer for observer-conditioned information. It formalizes how observer, transformation, uncertainty, provenance, equivalence classes and recoverability are represented without collapsing them into context-free values.

Production compiler, runtime and operating-system components may implement this work downstream, but the language specification and scientific reference model remain research-programme assets rather than venture ownership.

- Immediate research objective: identify which candidate primitives survive falsification strongly enough to deserve first-class representation.
- Implementation boundary: Invariant OS may implement the specification but is not itself counted as a foundational research programme.

3. AFC - ANNIHILATION-FREE COMPUTING

Research Programme III - new foundational research programme

AFC investigates whether computation can be constructed from relational distinctions, transformations and preserved invariants without treating zero as a primitive computational state. The programme does not require the ontological claim that zero 'does not exist'; it tests whether primitive-zero representation is computationally necessary.

The initial falsifiable target is to construct a computational substrate in which zero is never represented as a primitive value while demonstrating arithmetic, branching, memory and state-transition equivalence to a conventional machine.

- Research tracks: zero-primitive-free arithmetic, relational logic, non-zero-sentinel memory/state representation and execution architecture.
- Failure condition: If the proposed system necessarily reintroduces an equivalent primitive under another name, the strong AFC hypothesis is not supported.

4. SACRA - SYMBOLIC AUTHORITY, CONVERSION & RITUAL AGENCY

Research Programme IV - symbolic authority, ritual transformation, identity and agency

SACRA investigates how symbolic authority, religious and cultural rule systems, ritual practice and conversion-like transformations reorganize perceived agency and identity. It studies authority anchoring, repentance and commitment as state-transition mechanisms; tradition as persistence across agents and generations; and possession/exorcism narratives as cases of perceived external agency and ritualized agency reassignment.

The programme does not assume supernatural claims as given; instead it formalizes how symbolic systems, ritual acts and belief structures can stabilize or alter behavior, self-models and cross-observer interpretations. A key hypothesis is that some authority-anchored transitions exhibit hysteresis, lock-in or effectively irreversible state changes.

- Research tracks: authority anchoring, ritualized agency reassignment, conversion/repentance transitions and tradition as intergenerational persistence.
- Failure condition: if the proposed framework reduces entirely to relabeling without measurable changes in identity, agency or state persistence, the strong SACRA hypothesis is not supported.

FOUNDATIONAL PROGRAMME LEADERSHIP

RARI - Emirhan Yıldırım: Founder & Research Lead | Ayşe Günnaz: Co-builder, Bioengineering & Human Systems

OCIL / OCIR - Emirhan Yıldırım: Founder & Research Architect | Ayşe Günnaz: Co-builder, Biological Semantics

AFC - Emirhan Yıldırım: Founder & Research Lead | Ayşe Günnaz: Co-builder, Cross-disciplinary Research

SACRA - Ayşe Günnaz: Program Lead | Emirhan Yıldırım: Assistant Research Lead

EDUCATION & PROFESSIONAL EXPERIENCE

Formal degree, planned formal-credit expansion, and research / technical roles.

EDUCATION

Current

BSc Student — Occupational Therapy, Istanbul Medipol University, Istanbul

Entering Year 3. Training includes anatomy and neuroanatomy, activity and occupational analysis, ADLs, clinical measurement, motor function, rehabilitation technology and evidence-informed practice. Complementary technical work spans computational neuroscience, biomedical data analysis, medical AI, CAD/assistive-device design and reproducible research software.

2026–2028

Formal-credit systems / computing study pathway — planned

Planned university-credit expansion in applied computer science and engineering. Priority targets from the current credential plan include Engineering Mathematics 1–2, Data Structures & Algorithms with Python, Introduction to Operating Systems, Microprocessors, C Programming, Unix/Linux, Probability & Statistics, Discrete Mathematics, Fundamentals of Programming, and selected AI / cybersecurity / networking courses. These are planned targets and are not listed as completed coursework until enrollment and transcript evidence exist.

PROFESSIONAL & RESEARCH EXPERIENCE

2026–present

Founder & Project Director — Reality & Agency Research Initiative (RARI)

Founded and directs the long-horizon programme on neuroscientific reality, invariants of agency, agency-preserving transformations, observer/filter reconstruction and recoverability under perturbation. Fiscally sponsored by the Institute for Education, Research, and Scholarships (IFERS), a U.S. 501(c)(3) public charity, under a Model C Preapproved Grant Relationship executed in August 2026; programme management and project IP remain with the Initiative.

2026–present

Founder & Research Architect — Invariant Computing Program

Founding umbrella programme connecting RARI science with an observer-aware information model, language/IR, runtime, operating-system and future hardware architecture research. Current status is programme formation and technical specification rather than a completed language or computer platform.

Current

Founder / Independent Research Lead — Humanization Tech

Leads applied engineering and product-development work in multimodal biomedical representation, bounded AI, trustworthy data infrastructure, rehabilitation technology and deep-tech prototyping. Maintained separately from the fiscally sponsored RARI programme.

Current

Founder / Systems Architect — Invariant OS

Design-stage computing-systems venture for production implementations of the Invariant Computing stack: language/toolchain, compiler/IR, runtime, GNU/Linux research layer, storage/networking and future architecture/ISA work.

Current

Founder / Technical Architect — Geometrodynamic Computing

Design-stage spatial-computing venture for 3D/4D world representation, manifold/state-space systems, coordinate transforms, visualization, sensor fusion and persistent world models.

July 2026–present

Independent AI Evaluation Contributor (Project-Based) — Outlier AI, Remote

Completed onboarding and quality screening covering ASR transcription standards, non-speech event tagging, visual and multimodal analysis, and model-response quality review. Assignments remain project-dependent.

Current

Research & Technical Contributor — Parmer Academy / Parkinson-DBS Research Environment

Contributed to projects involving approximately 800 Parkinson's disease patients with pre/post-operative video and clinical material; supported data processing, encoder-side changes, markerless gait analysis, 3D DBS-coordinate modeling and predictive stimulation-parameter workflows. Architected and deployed fDBS Systems.

Current

Chief Technology Officer (CTO) — Hoshi no Umi Studio & Purpura Yakamoz

CTO for both creative ventures under Studio Head Ayşe Gümaz; leads technical architecture, production tooling, digital infrastructure, automation, AI-assisted workflows and cross-ecosystem integration.

Developed an exploratory framework for coding efficiency and population-matrix geometry in hippocampal and prefrontal splitter cells; executed Camber Cloud analyses identifying a 51.2 ms temporal precedence of trajectory-specific information in the hippocampal representation; secured ADNI access for future translational work.

RESEARCH SOFTWARE, MVPS & TECHNOLOGY DEVELOPMENT

Applied systems used as test beds for invariance, provenance, bounded adaptation, representation and recoverability.

Humanization AI / Bounded Policy-Evolution MVP Persistent cognitive-systems architecture with typed events, workspace, memory, self/world records, metacognitive signals, restricted executable policy generation, isolated evaluation, audit artifacts and rollback-compatible boundaries. Claims remain focused on error-aware review routing rather than consciousness or unrestricted self-modification.	Prometheus Core & Research Backend Auditable representation and file/data infrastructure supporting provenance, integrity, reversible processing, manifests and multimodal human/biological workflows. Internal technical testing reconstructed 47/47 files with SHA-256 verification and no missing, extra or mismatched files.
Prometheus MRI MVP Integrity-gated MRI capsule research prototype with Q8, Q16 and exact fallback, downstream task preservation, anomaly-ranking analysis and cryptographic provenance. Internal study reported $\sim 4.002\times$ float-array compression with minimal segmentation-behavior drift and a separately gated $\sim 8.165\times$ anomaly profile; no clinical-equivalence claim.	Prometheus RouterKey / MoE Adapter MVP Training-free event codec for compact, integrity-protected representation of mixture-of-experts router states. Reported 0.86% event density, 97.77% top-1 expert agreement, 99.16% top-8 overlap and confidence-gated dense fallback; full end-to-end acceleration remains unproven.
RhythmWalk / Autonomous RAS Research MVP Fixed-first personalized rhythmic auditory guidance informed by participant-held-out action ablation, cadence calibration, fail-closed quality checks, persistent audit logging and an optional bounded one-window slowdown probe. No treatment-benefit or adaptive-superiority claim.	Hardware-Aware Offline GLM System Active R&D for a locally operated GLM-family model environment intended to select execution, expert residency and fallback strategies according to RAM, storage, CPU/GPU resources and model metadata. Current validation is limited to router-state representation rather than full-runtime equivalence.
NCL Research API Proposed research-facing infrastructure for multimodal EEG, EMG, gait, imaging, clinical notes and research files, including common representation, missing-data control, data/model cards and research reporting.	Prometheus One World / GameKey Early Unreal Engine 4.27 game/simulation architecture under Elaveteos Studio. Defines an authoritative World State and Event Graph plus a Prometheus Interchange Representation (PIR) spanning human, activity/occupation, agent, household, object, physics, vehicle, settlement, production, economy, institution, knowledge, social and creator domains.
Observer-Conditioned Information Language / IR — working architecture Design-stage language and intermediate-representation work for observer, filter/transformation, uncertainty, provenance, equivalence class, recoverability, perturbation evidence and invariant relations. Not yet represented as a completed language standard.	Invariant OS — Generation-0 research system Planned GNU/Linux-based implementation layer for the new information model, compiler/runtime experiments, system-level metadata, storage and distributed observer-aware services before any custom-kernel or hardware claims.

UNIFYING TECHNICAL LOGIC

These systems are treated as applied test beds for structure-preserving transformation: preserving decision-relevant identity, downstream behavior, provenance, recoverability or bounded agency while representations, compression levels, policies, interfaces or execution conditions change.

CLINICAL, COLLABORATIVE & LEADERSHIP WORK

Translational work, external collaborations, recognition, funding experience and service.

CURRENT CLINICAL & ASSISTIVE-TECHNOLOGY DEVELOPMENT

- Personalized Passive Upper-Extremity Support for Children and Adolescents with Duchenne Muscular Dystrophy — manuscript/project in preparation. Integrates PUL 2.0, COPM, task performance, posture, fatigue, reach, user experience, engineering safety gates, SolidWorks-based parameterization and user-centered iteration.
- Adjacent translational pathway: analogous passive-support concepts are being considered for older adults, post-stroke users and age-related upper-extremity weakness; planned AI-assisted motion/requirement extraction remains unvalidated.

CURRENT COLLABORATION & PROJECT PIPELINE

- PRAISE Evidence Synthesis & Occupational Context Framework — proposed 8–12 week evidence-synthesis roadmap for possible TOP Lab / PRAISE work on whether sensor-detected pressure-relief events preserve enough occupational, environmental, temporal, behavioral and participation context for functional interpretation.
- Project operating principle: simpler adequate models remain the default; unmeasurable constructs are treated as research gaps; prospective extensions are not presented as established findings.

HONORS, SELECTIVE RECOGNITION & COMPETITIVE ADVANCEMENT

- | | |
|-------------|---|
| 2026 | The Hackers Conference 2026
First-Time Attendee invitation and roster inclusion after Invitations Committee review; scholarship award/coupon received for attendance support, while international airfare remains self-funded. |
| 2026 | Foresight Institute Vision Weekend US 2026
Accepted Volunteer Participant for the 13–15 November San Francisco Bay Area programme; programme access outside volunteer shifts, meals provided, travel/accommodation not funded. |
| 2026 | xAI
Applied Math and AI Evaluation: advanced to the second stage of a selective global vetting process through a CodeSignal assessment. |
| 2025 | TEKNOFEST
participated in 14 technology competitions; all submissions advanced beyond the first stage, with a semifinal result in Oncology 3T. |
| 2026 | Falling Walls Lab Budapest 2026 - Invited Attendee
Invitation to the 15 September invitation-only event by Tech In The City, the German Embassy, and HUN-REN Biological Research Centre; unable to attend due to Schengen visa constraints. |

GRANT APPLICATIONS & FUNDING EXPERIENCE

- TÜBİTAK 1001 proposal — research contribution within the Parmer environment.
- TÜSEB A2 proposal — preparation with the NEED / Humanization Tech team and additional Parkinson-related pilot work for international submission.
- Elaveteos Studio / Prometheus One World / GameKey — Epic MegaGrants identified as the primary 2026 funding route, contingent on a working Unreal vertical slice and measurable performance/provenance evidence; ID@Xbox and selected publisher pathways remain contingent follow-ons.
- Assistive technology — planned competition/validation pathways include Stanford Longevity Design Challenge, Microsoft Imagine Cup, James Dyson Award, Prototypes for Humanity, ASME ISHOW, MIT Solve and Global Student Prize; listed as targets, not awards.

PROFESSIONAL ASSOCIATIONS & ACADEMIC SERVICE

- | | |
|---------------------|---|
| 2025 | Board Member Responsible for Finance
Youth Commission, Turkish Occupational Therapy Association (WFOT member organization). |
| 2025–present | Class Representative
Department of Occupational Therapy, Istanbul Medipol University. |
| 2026 | Contributed to organization and on-site operations of an international occupational therapy congress. |
| 2026–present | Member
Cohere Labs Open Science Community; research interests directed toward ML Understanding and AI Safety & Alignment subgroups. Community affiliation, not employment, grant or fellowship. |

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

Recent and professionally relevant work is foregrounded; the full historical scholarly corpus is retained in appendices with explicit claim boundaries.

PEER-REVIEWED — SUBMITTED / UNDER REVIEW [2]

- 1. Yildirim E. Humanization Representational Theory (HuReT): A Falsifiable Geometric-Arithmetic Framework for Cognitive Computation. Sole author. Target/submission: Cognitive Computation. Zenodo DOI: 10.5281/zenodo.21382709. Reproducible proof-of-principle uses 1,797 open digit images and synthetic symmetry tasks; core findings support task-conditional geometry and transformation matching while explicitly not establishing biological universality.
- 2. Quality-Aware Markerless Video Assessment of Gait and Axial Impairment in Parkinson's Disease: A Retrospective Clinical Video Study. Co-author. Submitted to Movement Disorders, 16 July 2026. Contributed to a quality-aware markerless gait and axial-assessment pipeline using retrospective clinical video.

CURRENT PREPRINTS / ZENODO / SUBMISSION-READY MANUSCRIPTS [7]

- 1. Yildirim E. From Relational Invariants to Spectral Reconstructibility: A Four-Stage Falsification Study of Neural State Identity Under Perturbation. Research article / Zenodo publication, version 1.0.0, published 10 August 2026. Four-stage secondary analysis across two independent public animal datasets testing neural-state preservation under perturbation. PRI-v1, exactly gain-invariant RLG-v1, transformation-family Atlas-v1, and operator-valued QOID-v1 did not establish a generalizable identity invariant; the surviving exploratory signal motivates the prospective Spectral Reconstructibility Hypothesis. The study is explicitly preclinical and does not claim brain-death reversal, a clinical BD/DNC marker, consciousness, or physical neural quantumness.
- 2. Yildirim E. FORGE-L: A Permutation-Invariant Hypergraph Rewrite Model for Recursive Geometrogenesis - Embedding Fibonacci substitution in exchangeable local hypergraph dynamics. Original research article, preprint v1.0, 26 July 2026. Zenodo DOI: 10.5281/zenodo.21609625. The model establishes a concrete permutation-invariant hypergraph rewrite dynamics and reports a reproducible negative boundary: golden recursion, connectivity, and hub suppression are insufficient to establish physical 3+1-dimensional spacetime.
- 3. Yildirim E. FORGE-YM: Reflection-Positive Gauge Dynamics on Rewriting Hypergraphs and a Falsifiable Continuum Mass-Gap Conjecture - A constructive candidate framework for four-dimensional Yang-Mills theory. Preprint / submission version v0.3, 27 July 2026. Zenodo DOI: 10.5281/zenodo.21612528. The manuscript proves finite constructive statements and states the unresolved continuum mass-gap step as a falsifiable conjecture; it explicitly does not claim a completed Millennium Prize proof.
- 4. Safe Bounded Policy Evolution for Error-Aware Routing in Multimodal AI Systems. Humanization AI manuscript package; manuscript prepared, not yet publicly shared. Evaluated restricted review policies on 15,462 fixed predictions from 127 participants across EEG and wearable gait domains. Evolved routing improved over the bootstrap policy but did not outperform a simple uncertainty-only heuristic in aggregate.
- 5. Fixed-First Personalized Rhythmic Auditory Guidance Informed by Participant-Held-Out Action Ablation. RhythmWalk manuscript; manuscript prepared, not yet publicly shared. Harmonized 174,014 windows from 11 public gait-related sources; the core cue-response comparison used 99,595 transitions from 20 participants. Action-aware prediction did not show consistent incremental value, supporting a fixed-first rather than unsupported adaptive-superiority design.
- 6. Confidence-Gated Event Coding of Mixture-of-Experts Router States. Prometheus RouterKey manuscript; manuscript prepared, not yet publicly shared. Reported 78.05 bits/token at 0.86% event density, 97.77% top-1 agreement, 99.16% top-8 overlap, and 99.51% overall exact expert-set agreement with fallback; runtime acceleration remains to be demonstrated.
- 7. Integrity-Gated Adaptive MRI Capsules Preserve Downstream Task Behavior Under Four- and Eightfold Compression. Prometheus MRI original-article package; manuscript prepared, not yet publicly shared. Evaluated 484 primary brain-tumor cases and 1,936 MRI volumes; reported 4.002x float-array compression with maximal Dice drift below 0.0003 and a separately gated 8.165x anomaly-ranking profile, without claiming clinical equivalence or external predictive validation.

BOOKS & MONOGRAPHS — ZENODO SCHOLARLY CORPUS [7]

- 1. Yildirim E. THE MONOGRAPH of Somatic Information FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 2. Yildirim E. THE MONOGRAPH of THE HOLISTICITE FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 3. Yildirim E. THE MONOGRAPH of The Integrated Semiotic FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 4. Yildirim E. THE MONOGRAPH of THE MOVEMENT FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 5. Yildirim E. THE MONOGRAPH of THE OCCUPATION FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 6. Yildirim E. THE MONOGRAPH of THE REALITY FRAMEWORK. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.
- 7. Yildirim E. THE PARTICLE FRAMEWORK MONOGRAPH. Monograph / book-length theoretical work archived in the supplied Zenodo corpus.

CONFERENCE ACCEPTANCES, ABSTRACTS & POSTERS

- 1. International Parkinson and Movement Disorder Society (MDS) Congress 2026 - Accepted. Lead Asymmetry and Axial Symptom Response to Deep Brain Stimulation in Parkinson's Disease: A Synthetic Pilot Modeling Study. First Author and Lead Modeler. Synthetic pilot models reported $R^2 = 0.71$ and balanced accuracy = 0.79 for freezing-of-gait responder identification.
- 2. International Parkinson and Movement Disorder Society (MDS) Congress 2026 - Accepted. Video-Based Gait Analysis for Objective Assessment of Motor Severity and Surgical Outcome in Parkinson's Disease. Co-Author and Technical Contributor. Supported pose-

estimation and kinematic/spectral analyses reporting AUC = 0.89 for motor-severity discrimination and AUC = 0.82 for postoperative DBS improvement.

- 3. IAPRD 2026 - 31st World Congress on Parkinson's Disease and Related Disorders - Accepted. Axial Progression-Informed Modeling of Freezing of Gait and Balance Outcomes Following Deep Brain Stimulation. First Author and Lead Modeler. Three-tier longitudinal and synthetic framework reported AUC = 0.80, with medio-lateral lead asymmetry treated as an outcome modifier.

CURRENT RESEARCH-CORPUS POSITIONING

Earlier physics-, dimensionality-, computation-, language- and occupational-science manuscripts are retained as exploratory or historical branches unless independently validated. They are not treated as established physical theories. Within the revised CV architecture, several computation/language works are presented as early exploratory precursors to the current Invariant Computing programme rather than as independent validated paradigms.

PATENT-ORIENTED TECHNICAL DISCLOSURES

64 public technical-disclosure / patent-oriented manuscripts preserved in the Zenodo corpus. These are disclosure records, not granted or examined patents.

NEED — neuro / cognitive inference & modulation

- 1. A System and Method for the Geometrodynamical Inference and Modulation of Cognitive States.
- 2. A System and Method for Geometric State Inference and Somatic Feedback.
- 3. An Apparatus and Method for Geometrodynamical Neurohaptic Feedback via a General Reality Manifold.
- 4. Geosemantic Neurohaptic Interface for Brain-Computer Interaction.
- 5. System and Method for Geometric Inference of Neurophenomenological States and Somatic Feedback.
- 6. System and Method for Geometrodynamical Neuro-State Mapping via a Mobile Peripheral Device.
- 7. System and Method for Geometrodynamical Inference of Neurophenomenal States via Manifold Resonance Tomography.
- 8. A System and Method for Large-Scale Geometric Analysis of Aggregated Neurophenomenological Data.

SEED — spatiotemporal / manifold interaction & visualization

- 9. A Geometrodynamical System and Method for Spatiotemporal Analysis and Manipulation via a Recursive Manifold Architecture.
- 10. A System and Method for Geometrodynamical Reality Modeling and Interaction on a Recursive Fractal Manifold A System and Method for Geometrodynamical Reality Modeling and Interaction on a Recursive Fractal Manifold. ● Creators:.
- 11. A System and Method for Collaborative Geometrodynamical Visualization and Interaction in a High-Dimensional Manifold.
- 12. A System and Method for Active Geodesic Probing of a Spatiotemporal Manifold.
- 13. GEO-SEMANTIC INTERFACE: A System and Method for Navigating and Visualizing High-Dimensional Geometrodynamical Data.
- 14. A System and Method for Interactive Geodesic Navigation and Collaborative Analysis of High-Dimensional Manifold Data.
- 15. GEOMETRODYNAMICAL MANIFOLD SCANNING SYSTEM AND METHOD.

MEED — material / macro-scale state analysis & control

- 16. System and Method for Geometrodynamical State Control via a Recursive Manifold.
- 17. A System and Method for Geometrodynamical State-Space Analysis and Visualization on a Mobile Computational Device.
- 18. APPARATUS AND METHOD FOR GEOMETRODYNAMICAL MATERIAL STATE ANALYSIS VIA A MOBILE SENSING MODULE.
- 19. System and Method for Geometrodynamical Control of Material States via a Somatic Mixed-Reality Interface.
- 20. System and Method for Geometrodynamical State-Mapping via Macro-Scale Manifold Curvature Determination.
- 21. System and Method for Geometrodynamical Manifold Tomography via a Portable Apparatus.
- 22. A System and Method for Geometrodynamical Material Analysis and Manipulation in a Mixed Reality Environment.
- 23. System and Method for Geometrodynamical State-Mapping of Macroscopic Assets via a Satellite-Anchored Distributed Sensor Network.

FEED — organic matter / spectroscopy / metabolic / agriculture

- 24. A System and Method for Geometrodynamical State Analysis of Organic Matter.
- 25. System and Method for Geometrodynamical Analysis of Materials via Portable Spectroscopy.
- 26. System and Method for Geometrodynamical Mapping of Metabolic States using Accelerated Magnetic Resonance Spectroscopic Imaging.
- 27. A System and Method for Geometrodynamical Modeling and Prediction in Agricultural Systems.
- 28. AuraScan: A System and Method for In-Situ Geometrodynamical and Spectroscopic Analysis of Foodstuffs via a Mobile Platform.
- 29. System and Method for Geometrodynamical State-Space Analysis of an Analyte.
- 30. System and Method for Geometrodynamical Mapping of a Metabolic State-Space Manifold via Magnetic Resonance Spectroscopy.
- 31. A System and Method for Geometrodynamical Analysis of Agricultural Systems and Supply Chain Optimization via a Unified Multi-Dimensional Manifold.
- 32. A System and Method for Geometrodynamical State-Mapping of Physical Matter via Fractal Resonant Analysis and Classification Technology (FRACTech).

WEED — creative / physiological / satellite / immersive systems

- 33. A Geometrodynamical System and Method for Mapping and Modulating Cognitive Trajectories on a High-Dimensional Manifold.
- 34. A System and Method for Geometrodynamical Modeling and Navigation of a Creative State-Space.
- 35. A System and Method for Geometrodynamical State-Mapping via a Multi-Modal Physiological Sensor Apparatus.
- 36. System and Method for Geometrodynamical Navigation of Abstract Creative Manifolds in a Mixed Reality Environment.
- 37. System and Method for Geometrodynamical Routing in a Fractal Low Earth Orbit Satellite Constellation.

- 38. System and Method for Geometrodynamical Modeling and Visualization of a User's Cognitive-Creative State.
- 39. A System and Method for Geodesic Neuro-Cognitive Mapping in an Immersive Mixed Reality Environment.
- 40. A Geometrodynamical Satellite Network for High-Fidelity Symbiotic Data Streams.
- 41. System and Method for Geometrodynamical State-Mapping via a Multi-Modal Physiological Sensor Apparatus.

LED — guidance / optimization / haptics

- 42. A System and Method for Geometrodynamical Inference and Geodesic Guidance on a Generative Manifold.
- 43. A System and Method for Geometrodynamical State-Mapping and Proactive Haptic Guidance.
- 44. Apparatus and Method for Geometrodynamical State-Space Mapping and Proactive Haptic Guidance.
- 45. System and Method for Geodesic Guidance on a Dynamic State-Space Manifold for Human-System Performance Enhancement.
- 46. A System and Method for Geometrodynamical Analysis and Optimization of Large-Scale Systems via a Universal State-Space Manifold.
- 47. A System and Method for Geometrodynamical State-Mapping and Haptic Guidance on a High-Dimensional Manifold.
- 48. A Geometrodynamical Haptic Orthosis for State-Space Navigation and Motor Learning.
- 49. A System and Method for Geometrodynamical Analysis and Optimization of Large-Scale Systems on a Universal State-Space Manifold.

RED — intent / semiotics / orbital analytics / distributed sensing

- 50. A System and Method for Geometrodynamical Intent Formalization and Semiotically Grounded Artificial Intelligence.
- 51. A Method and System for Geometrodynamical Haptic Guidance Based on User Intent Manifolds.
- 52. A Geometrodynamical Haptic Interface System and Method for Mapping and Guiding User Intent on a High-Dimensional Manifold.
- 53. Geodesic Haptic Interface for Somatic State Navigation and Control.
- 54. Geometrodynamical System for Macro-Scale Orbital Analytics and Semiotic Data Generation.
- 55. System and Method for Geodesic Haptic Guidance Based on an Inferred Manifold of User Intent.
- 56. A System and Method for Geometrodynamical Anomaly Detection and Resource Allocation via a Distributed, Self-Organizing Sensor Manifold.

MSH — phenomenological / neural / biosensing / neuromodulation

- 57. System and Method for Geometrodynamical Modulation of a Subject's Phenomenological Manifold for Therapeutic Purposes.
- 58. SYSTEM AND METHOD FOR GEOMETRODYNAMICAL MODULATION OF A SUBJECT'S NEURAL STATE-SPACE MANIFOLD VIA A MOBILE COMPUTING DEVICE.
- 59. A Biometric Sensing Apparatus for Dynamic Modeling of a Subject's Phenomenological Manifold and a System for Therapeutic Geometric Modulation.
- 60. System and Method for Geometrodynamical Manifold Reconstruction via Magnetic Resonance.
- 61. A System and Method for Geometrodynamical Neuromodulation via a Constellation Waveform Emitter Array.
- 62. A System and Method for Geometrodynamical State-Space Reconstruction via a Mobile-Integrated Biosensing Platform.
- 63. System and Method for Geometrodynamical State-Space Mapping from Multimodal Physiological Time-Series Data.
- 64. A System and Method for Geometrodynamical Neuromodulation via a Constellated Emitter Platform.

HISTORICAL EXPLORATORY SCHOLARLY CORPUS

67 Zenodo preprints / theoretical papers / essays / technical research papers retained for completeness. Their presence documents intellectual development; it does not convert speculative branches into established science.

- 1. Yıldırım E. A Dynamical Systems Model of Occupational Engagement: The Principle of Volitional Perturbance.
- 2. Yıldırım E. A Formalization of Yıldırım's Geometry: A Novel Family of Zonotopes and their Application to M-Theory Compactification.
- 3. Yıldırım E. A Geometric Framework for Auditory Perception: The Octavector Model of Neural Sound Encoding.
- 4. Yıldırım E. A Volumetric Gluon Model: Re-evaluating Dimensionality in Quantum Chromodynamics.
- 5. Yıldırım E. Beyond the Turing Paradigm: Prometheus and the Architecture of Calculated Life Forms. Zenodo DOI: 10.5281/zenodo.18625555.
- 6. Yıldırım E. Causal Computation on Quintessence Manifolds: A Geometric Framework for Foundational Coding Logic.
- 7. Yıldırım E. Computational Singularity Theory: A Zero-Dimensional Framework for Code, Computation, and Artificial Intelligence.
- 8. Yıldırım E. Decimal Coordinate Theory (DCT): Reconceptualizing the Base-10 System as a Multidimensional Phonetic-Geometric Framework.
- 9. Yıldırım E. Five-Dimensional Orbital Field Theory: A Geometric Reinterpretation of the Standard Model.
- 10. Yıldırım E. General Reality Theory: A Geometric Framework for the Subjective Perception of High-Dimensional Information in Biological Systems.
- 11. Yıldırım E. General Realness Theory: An Eleven-Dimensional Framework for the Objective Representation of Perceived Information.
- 12. Yıldırım E. Geometric-Phonetic Number Theory (GPNT): A Unified Framework for Numerical Consciousness.
- 13. Yıldırım E. Geometrodynamics of the Soul: A Theory of Structural Isomorphism in Abrahamic Theology and Physics.
- 14. Yıldırım E. Glomerular Field Theory: A Bidimensional Framework for Olfactory Coding.
- 15. Yıldırım E. Graviton Dimensionality Unification (GDU): A Nona-Dimensional Framework for Quantum Gravity.
- 16. Yıldırım E. Hexadimensional Influence Theory: A Unified Framework for Computation.
- 17. Yıldırım E. Holographic Computation: A Framework for Two-Dimensional Structural Inputs in Algorithmic Systems.
- 18. Yıldırım E. Hyper-State Proprioception (HSP): A Five-Dimensional Framework for the Neuronal Encoding of Bodily State.
- 19. Yıldırım E. Mathematical Language and Grammar (MLG): A Geometrodynamical Unification of Semiotic Systems via The Reality Framework.
- 20. Yıldırım E. Mathematical Language and Grammar Theory (MLGT): A Unified Framework for the Foundation of All Languages.
- 21. Yıldırım E. Octadynamic Resonance Theory: An Eight-Dimensional Re-evaluation of the Standard Model's Fundamental Particles and Interactions.

- 22. Yıldırım E. Point-Matrix String Theory (PMST): A Preonic Substructure for Quarks from Zero-Dimensional Branes.
- 23. Yıldırım E. Predictive Adaptation through Occupational History (PATH): A Neuro-Computational Framework for Occupational Science.
- 24. Yıldırım E. Prometheus MVP Academic Validation Experiment – Phase 1: Theoretical Foundation Analysis.
- 25. Yıldırım E. Qualia Strand Theory: A Heptadimensional Framework for Information in Computational Systems.
- 26. Yıldırım E. Relativity Particles: A Unified Geometrodynamical Framework from Universal Dimensional Curvature.
- 27. Yıldırım E. Stochastic Filtration Theory (SFT): A Numerical-Phonetic Paradigm for Decoding Infinite Randomness.
- 28. Yıldırım E. Syntactic String Theory: A Unidimensional Framework for Computation.
- 29. Yıldırım E. The Archetypal Selves Framework for Occupational Engagement (ASFOE): An Anthropological and Neuroscientific Re-envisioning of Personhood in Occupational Therapy.
- 30. Yıldırım E. The Archetypal Theory of Socio-Occupational Engagement (ATSOE): A Neurosociological Model for Occupational Therapy.
- 31. Yıldırım E. The Autopoietic Model of Occupation: An Active Inference and Thermodynamic Framework for Occupational Science.
- 32. Yıldırım E. The Brane-Constituent Quark (BCQ) Hypothesis: A D1-Brane Model for Quark Substructure within M-Theory.
- 33. Yıldırım E. The Cyb-Occupational Systems Theory: A Neuro-Engineering Framework for Human Performance in the 21st Century.
- 34. Yıldırım E. The Digitalized of Cosmos: A Geometrodynamical Unification of Physics, Computation, and Consciousness.
- 35. Yıldırım E. The Dimensionality Postulate: A Ten-Dimensional Framework for Universal Knowledge Representation and Computation.
- 36. Yıldırım E. The Ennead-Vector Framework for Interoceptive Modulation of Neuronal Computation.
- 37. Yıldırım E. The Enneastuctural Model of Occupational Engagement (EMOE): A Neuro-Phenomenological Framework for Occupational Therapy.
- 38. Yıldırım E. The General Occupation Theory: An Eleven-Dimensional Framework for Non-Individual Agents.
- 39. Yıldırım E. The Geometric Theory of Neuronal Processing: A Tensorial Framework for Magnetoceptive Integration.
- 40. Yıldırım E. The Geometrodynamical Origin of Particles: A Black Hole-Brane Unification Model.
- 41. Yıldırım E. The Heptadic Framework for Neuro-computational Occupational Dynamics (H-NOD): A Synthesis of Occupational Science and Complex Systems Theory.
- 42. Yıldırım E. The Holisticite Framework: A Geometric Architecture for Embodied Intelligence and Neuro-computational Systems.
- 43. Yıldırım E. The Holo-Manifold Theory of Vision: A Seven-Dimensional Framework for Neural Processing.
- 44. Yıldırım E. The Hyper-Lepton Framework: A Geometric Origin for Fermion Generations.
- 45. Yıldırım E. The Integrated Semiotic Framework (ISF): A Geometric Synthesis of Objective Reality and Subjective Experience for Embodied Artificial Intelligence.
- 46. Yıldırım E. The Intersubjective Contemplative Experiment (ICE): A Neuro-Phenomenological Framework for Investigating Shared Experience via Pure Intentionality.
- 47. Yıldırım E. The Intrinsic Dimensionality Framework: A Six-Dimensional Re-interpretation of Bosons in the Standard Model.
- 48. Yıldırım E. The Movement Framework (TMF): A Geometrodynamical Unification of Occupation and Emergence.
- 49. Yıldırım E. The Nociceptive Singularity: A Theory of Pain as an Information-Driven Phase Transition in a Critical System.
- 50. Yıldırım E. The Nona-Dimensional State Formalism: A New Foundation for Computational State Representation.
- 51. Yıldırım E. The Occupation Framework (TOF): A Unified Geometric Model for the Analysis of Human and Systematic Efficiency.
- 52. Yıldırım E. The Octave-State Formalism: A State-Space Representation of Computational Loops.
- 53. Yıldırım E. The Particle Framework (TPF): A Geometrodynamical Approach to Emergent Matter and its Implications for Materials Engineering.
- 54. Yıldırım E. The Reality Framework: A Geometrodynamical Model of the Constituted Perception-Action Loop.
- 55. Yıldırım E. The Reality Law: A Geometrodynamical Model of the Perception-Action Loop.
- 56. Yıldırım E. The Recursive Universe: A Geometrodynamical Framework for a Theory of Everything.
- 57. Yıldırım E. The Somatic Information Framework: A Geometric Realization of Sensorimotor Contingencies on the General Reality Manifold.
- 58. Yıldırım E. The Spatiotemporal Information Framework (STIF): A Relativistic Model of Computation.
- 59. Yıldırım E. The Spatiotemporal Tensor Model of Gustatory Perception.
- 60. Yıldırım E. The Theory of Intrinsic Gauge Dimensionality.
- 61. Yıldırım E. The Theory of Relational Occupation Dynamics (ROD): A Neurobiological Field Model of Dyadic Engagement.
- 62. Yıldırım E. The Theory of Semiotic Occupation (TSO): A Neuro-Linguistic Framework for Zero-Individual Occupations.
- 63. Yıldırım E. The Thermostate Manifold: A Ten-Dimensional State-Space Model of Neuronal Thermoreception.
- 64. Yıldırım E. The Triadic Occupational Nexus (TONE) Theory: A Complex Systems Approach to Neuro-Occupation.
- 65. Yıldırım E. The Volumetric Lambda Calculus: A Holographic Formalism for State-Space Computation.
- 66. Yıldırım E. Unidimensional Tactile Field Theory: A Dimensionality Reduction Model of Somatosensation.
- 67. Yıldırım E. Vestibular Spacetime Tensor (VST) Theory: A Geometric Framework for Neural Computation and Motor Control.

SUPPORTING MATHEMATICAL, SUPPLEMENTARY & REPRODUCIBILITY MATERIALS

The archived corpus also contains 52 appendix, mathematical-proof, validation, duplicate-version or companion files associated with the main scholarly works. These are treated as supporting research objects rather than inflated into separate publication counts.

TECHNICAL, CLINICAL & RESEARCH SKILLS

Current skill profile aligned with the revised Invariant Computing / RARI / systems architecture.

Programming & data

Python; PyTorch; scikit-learn; pandas; grouped and nested cross-validation; leave-one-animal/participant-out evaluation; permutation testing; bootstrap uncertainty; falsification-driven analysis; reproducible pipelines; scientific writing and structured evidence synthesis.

Systems & computing

Linux/Unix-oriented systems work; provenance and integrity; reversible transformation; bounded code generation; rollback concepts; resource-aware execution; canonical world/event graphs; domain-IR schema design; persistent-world architecture; operating-system and processor/ISA study pathway.

Neural identity & geometry

Representational geometry; Gram matrices; CKA/RDM concepts; covariance/eigenvalue spectra; entropy/purity/effective rank; spectral-capacity analysis; operator-valued embeddings; perturbational analysis; recovery prediction; ROC-AUC, AUPRC and log loss.

AI, imaging & signals

Computer vision; medical video; MRI representation; EEG and wearable-sensing concepts; model prototyping; clinical-correlation analysis; human-in-the-loop evaluation; ASR standards; multimodal review.

Spatial / 3D / engineering

3D DBS-coordinate modeling; SolidWorks-linked CAD workflows; geometrodynamics/manifold visualization concepts; Unreal Engine 4.27 architecture; persistent-world and interchange-representation design.

Occupational therapy & rehabilitation

Occupational profiling; activity and ADL analysis; occupational-context measurement; pediatric and neurological rehabilitation interests; clinical measurement; assistive technology; user-centered fitting and rehabilitation-device concepts.

RESEARCH & PROFESSIONAL INTERESTS

Invariant computing; observer-aware computation; programming languages and intermediate representations; operating systems and computer architecture; neuroscientific reality; invariants of agency; agency-preserving transformations; perturbation-driven identifiability; neural-state identity and recoverability; representational geometry; spatial/geometrodynamics computing; 3D/4D world representation; persistent-world simulation; scientific machine learning; bounded adaptive systems; trustworthy data infrastructure; medical AI; Parkinson's disease and DBS; gait and rhythmic auditory guidance; MRI; EEG; neurotechnology; rehabilitation engineering; haptics; human-computer interfaces; creative technology and theory-informed interdisciplinary research.

CREATIVE, MUSICAL & INTELLECTUAL PRACTICE

Retained as a separate personal-practice section rather than mixed with scientific achievements.

- 1. Violin - Approximately five years of playing experience; returning to structured technical and repertoire practice after interruptions.
- 2. Loveband - collaborative music project in formation, framed as an emerging violin-led dual-vocal pop-rock act. Current role: co-vocal, violin, songwriting/composition, and creative direction. Development priorities are original demos, a joint performance identity, and a small live repertoire; no commercial-release, touring, or funding-award claim is made.
- 3. Piano and voice - developing piano and vocal technique; has recorded early original amateur songs and is developing composition, performance, and digital-production practice.
- 4. Webtoon / animation development - collaborative narrative/visual project in development. Current planning includes a 10-12 episode first-season structure, character and relationship files, a pilot/one-shot, English-language text, visual model sheets by the collaborating artist, and a possible 6-10 minute short-animation / series-pitch pathway. Existing independent writing and artwork remain separately owned; jointly created characters, series identity, and adaptation rights are planned as joint IP. No publication, prize, or production-completion claim is made yet.
- 5. Poetry and fiction - writes poetry and short fiction, including an award-winning story; interested in postmodern and situation-centered narratives, memory, identity, fragmented perspective, and multilingual writing in English, German, French, Japanese, and Spanish as proficiency develops.
- 6. Chess - played in childhood and later participated in a simultaneous exhibition against a FIDE Master; returning through structured study and competitive-development goals.

EVIDENCE & CLAIM BOUNDARIES

Status labeling is a deliberate feature of the CV.

Early-stage software, theories, technical disclosures, manuscripts, design-stage ventures, prototypes, collaborations and funding plans are labeled by status. Zenodo deposits are research objects/preprints rather than peer-reviewed publications; technical disclosures are not granted patents; planned applications are not submissions or awards. RARI observer/filter, Markov-blanket-like boundary, higher-dimensional and quantum-inspired constructs remain research hypotheses or modeling devices rather than established neuroscientific facts. Day-1 and Day-2 synthetic experiments generated authoritative trajectories, while P1-P5 metrics were derived through frozen post-run analysis. The current evidence does not establish observer-independent reality reconstruction, biological universality, physical neural quantum effects, consciousness, immortality or a validated irreversibility threshold. Invariant Computing, Invariant OS and Geometrodynamics Computing are described according to their present programme/design-stage status; future incorporation, IP allocation and product validation require separate formalization.